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2027 CYCLE

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CUET UG

Biology

Organisms and Populations

Official Syllabus Unit X · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit X)	Skills the questions test
Population attributes — birth and death rate, age pyramids	Population attributes – birth and death rate, age pyramids.	Computing per-capita birth and death rates, reading the three pyramid shapes, judging population size from % cover or biomass.
Population growth — exponential and logistic growth	Population growth – exponential and logistic growth.	Deriving N_t from N_0, r and t; locating K; spotting where logistic growth runs fastest.
Life history variation	Life history variation.	Comparing breed-once with breed-many species, clutch size trade-offs, age at first breeding.
Population interactions — predation, competition, parasitism, commensalism, mutualism and amensalism	Population interactions – predation, competition, parasitism, commensalism, mutualism and amensalism.	Naming the interaction from its sign pair (+/+, +/−, +/−, +/0, −/0, −/−) and matching each type to its classic example.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit X opens with individuals and populations: how the environment squeezes them, how their numbers move, and how species treat each other.

Ecology and the Big Four Abiotic Factors

Why can a mango tree live in Delhi but not in London? Why does a snow leopard stay on cold mountains while a camel owns the desert? Ecology answers questions like these with one short idea: every organism is built for a particular set of surroundings, and the surroundings decide who survives where. This section builds the ladder of ecology, then meets the four big physical dials that control life everywhere — temperature, water, light and soil.

- **Ecology:** the branch of biology that studies how organisms interact with each other and with their physical surroundings.
- **Organism:** one single living individual — one banyan tree, one tiger.
- **Population:** all the individuals of **one species** living in **one area** at **one time** — every peacock in a forest, counted together.
- **Community:** all the populations of **different species** that share that area — peacocks, deer, grasses and trees taken together.
- **Biome:** a very large region with one broad climate and one characteristic life-form — desert, rain forest, tundra.

Notice the ladder: individuals build populations, populations build communities, and whole climate zones carry biomes. Every level up, ecology adds one new question — how many, of whom, and who is winning.

The four dials that run every habitat

Think of a habitat as a room with four control dials. Turn any dial too far and someone in the room dies. Ecologists group the physical surroundings into exactly these four big factors.

Factor	What it is	Who it limits — one example
Temperature	How hot or cold the place gets, hour by hour and season by season.	Nearly everyone. Mango trees cannot bear temperate winters; snow leopards cannot bear plains summers.
Water	How much water is available, and how salty it is.	Freshwater fish die in the salty sea; sea fish collapse in fresh rivers.
Light	How much sunlight falls, and for how many hours each day.	Plants that flower only on a fixed day-length schedule; all life in the pitch-dark deep sea.
Soil	The nature of the ground — texture, minerals, water-holding.	Plants rooted in it; the deer follow the grass that the soil can or cannot grow.

Keep a simple reading habit for every factor: **what it is → who it limits → one example**. Questions rarely ask the definition alone; they pair a factor with an organism and ask whether the pair fits.

Temperature: the master dial

Temperature on Earth swings from below zero at the poles to over 50 °C in summer deserts — and every organism carries its own range of tolerance.

- **Eurythermal:** organisms that tolerate a **wide** range of temperatures — humans, most mammals. “Eury” means wide.
- **Stenothermal:** organisms that tolerate only a **narrow** band of temperatures — many corals, polar fish. “Steno” means narrow.

- **Hot extremes:** some bacteria and archaea (ancient single-celled microbes) live in boiling hot springs — heat is their normal.
- **Cold extremes:** snow-dwelling animals manage life below zero with thick coats and steady body chemistry.

The two "-thermal" words are a classic matching pair: wide range, eurythermal; narrow range, stenothermal. Do not let a question swap them.

EURYTHERMAL VS STENOTHERMAL — THE STUDYING STUDENT

One student can study anywhere — a noisy hostel, a canteen, a silent library. Another needs one exact quiet corner or nothing happens. Both study; only the first tolerates a **wide range** of conditions. That is the whole mapping: the flexible student is **eurythermal**, the fussy one is **stenothermal** — and the "conditions" are degrees Celsius, not noise.

Water: the salt dial

Water matters twice — first how much of it exists, then how salty it is.

- **Salinity:** the amount of dissolved salt, measured in parts per thousand (‰).
- **Inland fresh waters:** salinity below about **5‰** — ponds, rivers, most lakes.
- **The sea:** salinity sits between **30‰ and 35‰** — over 30 parts of salt per thousand parts of water.
- **Halophiles:** "salt-loving" organisms that tolerate high salinity — they thrive in salty water where ordinary freshwater life dies out.

This is why the same fish never fills both a river and the sea: each body of water has its own salt level, and bodies are tuned to one level, not both.

Light: the timing dial

Light does two jobs — it powers photosynthesis, and it works as a clock.

- **Photoperiod:** the length of daylight in a day. Many plants flower only when the photoperiod reaches the right length, so their blooming follows the calendar.
- **Migration timing:** lengthening or shortening days cue birds and fish to start their seasonal journeys at the right time.
- **Deep-sea life:** sunlight never reaches the ocean floor, yet life exists there — it runs on food drifting down from the sunlit layers far above.

Soil: the ground dial

- **Texture and composition:** how much sand, silt and clay a soil holds, plus its minerals and water-holding capacity, decide **which plants** can root there.
- **Vegetation decides animals:** the animals follow the plants — no grass, no grazing deer, and no predators that hunt deer.
- **Aquatic sediments:** in water bodies the same logic runs on the bottom — the nature of the sediment decides which bottom-dwellers settle.

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One closing thought before we move on: these four factors rarely act alone. A desert is hot **and** dry; a mountain stream is cold **and** fast-flowing. Habitats squeeze organisms with several dials at once — which is exactly why organisms need strategies to cope, the subject of the next section.

How Organisms Cope: Responses and Adaptations

A desert afternoon touches 48 °C; a Himalayan night drops to −15 °C. No cell, no enzyme, no egg survives such swings happily — inside every organism there is a narrow band of conditions that actually works. So the real question is not "how hot is it outside?" but "what does the organism do about the outside?" Biology gives exactly four answers. Learn them as a set; the question will hand you a situation and expect you to name the right one.

The four responses

- **Regulate:** the organism keeps its inside steady no matter what happens outside — this steady state is called **homeostasis**. Humans sweat on a hot day and shiver on a cold one, and the body temperature stays near 37 °C either way. All birds and mammals regulate.
- **Conform:** the organism does nothing to fight the outside — its body temperature and chemistry simply follow the surroundings. Most invertebrates (animals without backbones) conform. Why not regulate? Holding a constant inside costs enormous energy, so small-bodied creatures skip the bill.
- **Migrate:** the organism cannot fix the conditions, so it **moves away** — temporarily. Siberian birds fly thousands of kilometres to Bharatpur in Rajasthan every winter, then fly back when home turns kind again.
- **Suspend:** the organism pauses life itself until better days. **Hibernation** is the winter pause (bears and ground squirrels sleep through the cold), **aestivation** is the summer pause (snails shut down in the heat), and **diapause** is a developmental pause used by many insects — even their eggs and pupae can hold time.

Watch the logic ladder: regulate is **fight**, conform is **give in**, migrate is **leave**, suspend is **pause**. Four verbs, four survival styles.

Response	What it means	Classic example
Regulate	Body keeps temperature and chemistry constant against outside swings (homeostasis).	Humans sweat at 45 °C and shiver at 5 °C — body stays near 37 °C.
Conform	Body state follows the outside; no energy spent holding a line.	Most invertebrates — a snail's body tracks the water around it.
Migrate	Move temporarily to kinder conditions, return later.	Siberian cranes wintering in Bharatpur, Rajasthan.
Suspend	Pause life processes until the season turns.	Hibernation (winter), aestivation (summer), diapause (insect eggs/pupae).

REGULATOR VS CONFORMER — HOUSE AND TENT

A house with an air-conditioner thermostat holds 24 °C whether the sun burns or the rain pours — it **spends electricity** to buy stability. A tent is 38 °C at noon and 20 °C at midnight — it **follows the weather** and spends nothing. The mapping is exact: the thermostat house is a **regulator** (constant inside, constant energy cost), the tent is a **conformer** (inside equals outside). A migratory family that drives to the hills for summer is the **migrator**, and a family that simply naps through the hot hours is **suspension**.

Adaptations: the body's long-term answers

Responses are hourly decisions; **adaptations** are body features built over generations — any attribute that helps an organism survive and reproduce in its habitat.

- **Desert plants — beat the water bill:** a **thick cuticle** (waxy outer skin) seals moisture in, leaves shrink into **spines** (in **Opuntia**, the flat green stem takes over photosynthesis), and stomata — the leaf's breathing pores — open **only at night** (the CAM pattern), so water vapour escapes in the cool hours instead of the burning day.
- **Allen's rule — shape follows climate:** mammals of colder regions carry **shorter ears and shorter limbs** than their warm-region relatives, because small sticking-out parts lose body heat fastest. Compare the compact Arctic fox with the long-eared desert fox.
- **Altitude — thicker blood:** people living at high altitude, where oxygen is thin, carry **more red blood cells (RBCs) and more haemoglobin** per drop of blood, so each litre of blood grabs more of the little oxygen available.
- **Archaea in hot springs:** these microbes run their chemistry in near-boiling water — their enzymes simply do not fall apart at temperatures that cook most life.
- **Desert lizard — the bask-and-burrow routine:** it lies in the morning sun to warm up, then burrows into cool sand when noon gets lethal. Its body follows the temperature — a conformer choosing its micro-climate by the hour.

Adaptation	Where it appears	What it achieves
Thick cuticle + spiny leaves + night-opening stomata (CAM)	Desert plants (**Opuntia**)	Cuts water loss; the stem photosynthesises while the spines stop animals from eating the water store.
Shorter ears and limbs	Cold-region mammals (Allen's rule)	Less surface area sticks out, so less body heat escapes into the freezing air.
More RBCs and haemoglobin	People at high altitude	Blood picks up enough oxygen from thin mountain air.
Heat-stable enzymes	Archaea in thermal springs	Chemistry keeps running in water that would boil other cells.
Bask by day, burrow at noon	Desert lizard	Conformer behaviour that rides out the worst hour of the day.

WORKED EXAMPLE — SPOT THE RESPONSE

Name the response in each situation, then check yourself:

- **A dog pants heavily in May: regulate** — the body is actively pushing heat out to hold its own temperature.
- **A jellyfish warms and cools with the sea: conform** — the inside follows the outside, no defence offered.
- **Flamingos arrive at Sambhar Lake in November: migrate** — the escape is temporary and ends with a return journey.
- **A snail sleeps through the summer inside its shell: suspend** (aestivation) — life is paused, not relocated.

One more habit to build: pause the answer and ask **is the inside steady?** Yes → regulate. No → is it moving, or paused, or just following? You have the vocabulary now.

Population Attributes

Suppose you are handed a forest and one page of a report. What numbers would make that forest's peacock population come alive on paper? Ecologists keep a small set of measurements — called **population attributes** — that describe any population the way a report card describes a student. Four of them matter for your syllabus: birth rate, death rate, sex ratio and age structure.

What exactly is a population?

- **Same species:** every member can potentially interbreed with the others — one reproductive club, not two.
- **Same area:** they share one defined place at one defined time.
- **Interbreeding is the test:** a city's dogs and a village's dogs may be the same species, but each group is its own population because the two groups do not mix.

Birth rate and death rate: count per head

- **Per-capita:** both rates are written **per individual**, not as raw totals — that is what "per capita" (per head) means.
- **Birth rate:** the number of births divided by the population size, per unit time. A rate of 0.02 means 2 births per 100 individuals each year.
- **Death rate:** the same arithmetic for deaths — deaths divided by population size, per unit time.

WORKED EXAMPLE — READING A POND'S REPORT CARD

A pond holds **2,000 fish**. In one year, 60 fry hatch and 40 fish die.

- **Birth rate** = 60 ÷ 2,000 = **0.03 per year** — three births per 100 fish.
- **Death rate** = 40 ÷ 2,000 = **0.02 per year** — two deaths per 100 fish.
- **Verdict:** births outpace deaths (0.03 > 0.02), so the population **grows**.

Now the trap-check: the same 60 births in a pond of 12,000 fish would give 60 ÷ 12,000 = 0.005 — a far **weaker** birth rate. Raw counts mislead; per-head rates tell the truth.

Sex ratio and age structure

- **Sex ratio:** the share of females to males in the population. India's 2011 census counted **943 females per 1,000 males** in humans — official numbers, and a reminder that real ratios rarely sit at a neat 1:1.
- **Age structure:** ecologists sort every individual into three buckets — **pre-reproductive** (too young), **reproductive** (breeding now) and **post-reproductive** (past breeding).
- **Age pyramid:** plot the percentage of individuals against these age groups and you get a pyramid whose **shape** announces the population's future.

Pyramid shape	What it looks like	What it means for the future
Growing	Wide base, narrow top — many children, fewer elders.	Births flood in faster than deaths; the crowd ahead is large (many developing countries).
Stable	Even sides — pre-reproductive, reproductive and post-reproductive groups match.	Births and deaths balance; the population holds level (moderate countries).
Declining	Narrow base, broad middle — fewer children than parents.	Births lag behind deaths; the population will shrink (a few developed countries).

Read the base first, always: the base holds tomorrow's breeders. A wide base promises a boom; a narrow base whispers a goodbye.

How big is a population? Three honest measures

- **Count the individuals:** the obvious method — 40 tigers, 2,000 fish. Works when individuals are big, visible and slow.
- **Percentage cover or biomass:** when counting fails, measure differently. A huge banyan tree is **one** individual but covers a whole field — ecologists use **per cent cover**. Grasses are too many to count — so weigh their **biomass** (total dry weight) instead.
- **Density:** the number of individuals squeezed into a unit area or volume — 40 tigers **per 100 km²** says more than "40 tigers" alone.

Attribute	What it measures	Typical form
Birth rate	New individuals per head, per unit time.	0.03 per year = 3 per 100 each year
Death rate	Deaths per head, per unit time.	0.02 per year = 2 per 100 each year

Attribute	What it measures	Typical form
Sex ratio	Females against males.	943 : 1,000 (India, 2011 census)
Age structure	Split across pre-reproductive, reproductive, post-reproductive.	Pyramid: growing, stable, declining
Size / density	Individuals, or their cover or biomass, per unit space.	40 tigers per 100 km²; % cover of banyan

Growth Models: Two Ways to Grow

Give a population endless food and space and it explodes; give it a real habitat with a real ceiling and it settles. Biology writes both stories as two growth models, and Figure 1 shows them side by side. Examine the shapes first — one curve shoots like a rocket (J), the other climbs, bends and flattens (S). Every formula in this section is just a sentence about one of these two shapes.

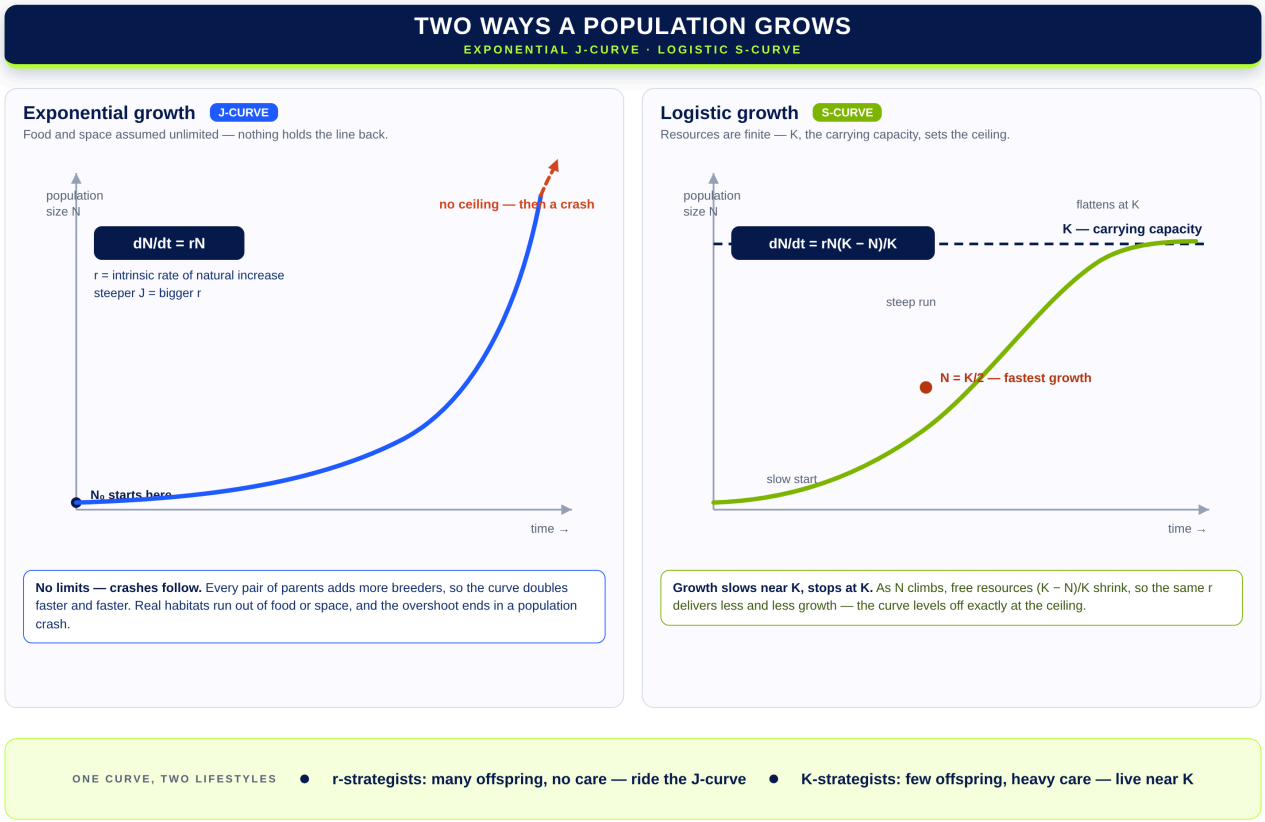


Figure 1: Two ways a population grows — the exponential J-curve with no ceiling, and the logistic S-curve capped by K, the carrying capacity.

Exponential growth: the J-curve

- **The assumption:** food and space are **unlimited** — the habitat never pushes back.
- **The mechanism:** every pair of breeders leaves more breeders, so the growth rate itself grows with the population — small crowd, slow rise; big crowd, vertical climb.

- **The shape:** plotted against time, N hugs the floor, then bends upward — a J. Nothing stops it, which is why real overshoots end in crashes.

EXPONENTIAL GROWTH — THE J-CURVE

$$\frac{dN_t}{dt} = rN_t$$

$N_t = N_0 \times e^{rt}$

Letter meanings, one pair per line:

- N_t = population size after time t
- N_0 = population size at the start
- r = intrinsic rate of natural increase (per-capita births minus deaths)
- e = the constant 2.718, the base of natural growth
- t = time elapsed

Read the top line aloud: "the change in population per unit time equals the per-capita rate times the current population." Bigger crowd or bigger r — either one speeds the growth.

- **How big can r be?** The maximum r recorded for the Norway rat is about **0.015** per unit time, and for the flour beetle about **0.12** — small numbers, huge consequences over many generations.

WORKED EXAMPLE 1 — EXPONENTIAL ARITHMETIC, EVERY STEP

A tank starts with **100 organisms**. They grow exponentially with $r = 0.5$ per unit time. How many organisms after $t = 2$ time units?

- **Step 1 — write the model:** $N_t = N_0 \times e^{rt}$
- **Step 2 — substitute the numbers:** $N_t = 100 \times e^{0.5 \times 2}$
- **Step 3 — tidy the exponent:** $N_t = 100 \times e^{1}$
- **Step 4 — use $e = 2.718$:** $N_t = 100 \times 2.718$
- **Step 5 — multiply:** $N_t = 271.8$

Rounding to whole organisms: about **272 individuals** after two time units. Notice the whole answer hangs on e^{rt}: get the exponent to $r \times t = 1$ and you are simply multiplying by 2.718.

Logistic growth: the S-curve

- **The reality:** no habitat is infinite. Sooner or later food, space or water runs short, and crowded populations feel it first.
- **Carrying capacity (K):** the maximum population size the habitat's resources can support for the long run. K is a property of the **habitat** — richer habitat, higher K.
- **The brake:** the fraction $(K - N_t)/K$ is the share of resources still free. When N_t is tiny, almost all of K is free and growth is nearly exponential; as N_t climbs toward K, the free share shrinks toward zero.
- **The shape:** slow start, steep middle run, then a flattening — an S that levels off **at** K, never past it.

LOGISTIC GROWTH — THE S-CURVE

$$\frac{dN_t}{dt} = rN_t \times \frac{K - N_t}{K}$$

Letter meanings, one pair per line:

- N_t = population size at time t
- r = intrinsic rate of natural increase
- K = carrying capacity — the habitat's long-run ceiling
- $(K - N_t)/K$ = fraction of the resources still unused

Read the top line aloud: "growth equals the exponential engine rN_t, multiplied by how much room is left." At $N_t = K$ the room is zero and growth stops; at $N_t = K/2$ half the room is left and growth runs fastest.

WORKED EXAMPLE 2 — LOGISTIC LOGIC: WHY FASTEST AT K/2

Question: a population sits at exactly $N = K/2$. Why is it adding individuals at the fastest possible rate?

- **Ingredient 1 — many breeders:** at $K/2$ the population is already half its ceiling, so a large crowd exists to produce young. Exponential engine rN_t runs loud.
- **Ingredient 2 — resources still free:** $(K - N_t)/K = (K - K/2)/K = 1/2$ — half the habitat's resources are still unused, so the brake is mild.
- **Compare the ends:** near $N = 0$ the engine is quiet (few breeders); near $N = K$ the brake is total (no free resources). The product peaks in the middle.
- **Verdict:** fastest growth sits at $N = K/2$ — not at K (growth has **stopped** there) and not at zero.

Exam habit: whenever a question mentions "maximum growth rate", place the population at half the carrying capacity before you touch anything else.

Two life-strategies: r and K

- **r-strategists:** small, short-lived species that pour energy into **many offspring with no parental care** — they exploit the J-curve side of life and bounce back fast after crashes.
- **K-strategists:** large, long-lived species that invest in **few offspring with heavy parental care** — they live crowded near K, where competition decides everything.
- **Why the names:** the letters are lifted straight from the equations — r is the growth rate, K is the ceiling. A species' strategy tells you which part of which curve it plays on.

Life history variation: every species writes its own plan

- **Breed once or many times?** Some species breed a single time and finish — certain bamboos flower once after decades of waiting, and the Pacific salmon spawns once, then dies. Most birds and mammals breed repeatedly across their lives.
- **Age at first breeding:** species under heavy danger of dying young start early; species that can afford to wait, wait. The same pressure writes different schedules in different habitats.

- **Clutch size — the offspring bet:** a **few-precious** plan produces a small number of large, protected young (birds that guard a handful of eggs); a **many-cheap** plan floods the water with thousands of tiny eggs (most fish) and lets chance sort it out.
- **The trade-off rule:** parents have one energy budget. Energy spent on numbers cannot be spent on care, and the reverse. Every life history is a compromise written by that budget.

Interactions I: Predation and Competition

No population lives alone. Rabbits meet foxes, acacias meet giraffes, and every meal is somebody's death. Ecologists call these species-vs-species deals **population interactions**, and they grade every deal with signs: + for benefit, − for harm, 0 for no effect. This section covers the two interactions where the signs turn brutal — predation and competition.

Predation: the energy pipeline

- **The deal:** the predator (+) kills and eats; the prey (−) pays with its life. A tiger takes a deer; a sparrow takes a seed — plants are prey too.
- **Energy flows upward:** predation is how the energy stored in plants and herbivores travels up to bigger hunters. Every food chain is a pipeline of predations.
- **Predators keep the brakes on:** without predators, herbivore populations multiply, strip the vegetation, and collapse their own habitat. Predators are the ecosystem's population control.
- **Biological control — using a predator on purpose:** when the prickly pear cactus spread across Australia and choked farmland, ecologists introduced a moth (**Cactoblastis**) whose caterpillars eat the cactus — the weed was brought under control without a drop of pesticide.
- **How prey fights back:**
  - **Camouflage:** insects shaped and coloured like dry leaves or twigs vanish against the branch they sit on.
  - **Toxins:** the milkweed plant **Calotropis** brews poisonous cardiac glycosides; the monarch butterfly stores milkweed poison and becomes distasteful to birds.
  - **Thorns and spines:** Acacia trees arm their stems so browsers pay in blood.
- **How predators answer:** speed, ambush, or team hunting — the pipeline keeps evolving at both ends.

Competition: the fight over a shared table

- **The deal:** both species lose (− and −), because both spend energy chasing the same limited food, light or space.
- **Intra-specific:** competition **within** one species — two peacocks for one mate.
- **Inter-specific:** competition **between different** species — two barnacle species for one rock face.
- **Competitive exclusion principle:** two species competing for the **exact same niche** (the full set of resources and roles a species uses) cannot coexist for long — the weaker competitor is eliminated.
- **Gause's experiment:** G. F. Gause grew two **Paramecium** (single-celled pond) species separately — both thrived. Grown together, sharing the same niche, one species (**P.**

aurelia) outgrew and **eliminated** the other (**P. caudatum**). Same niche, same table, one chair.

- **Resource partitioning — divide and survive:** species that split the niche can coexist. Five North American warbler species feed in **one** tree, but each species forages in its own zone — tips, middle branches, trunk — so five tables run in one tree.
- **Competitive release:** remove the competitor and the loser suddenly expands. J. H. Connell studied two barnacles on Scottish rocks: the larger **Balanus** crowded the smaller **Chthamalus** off the lower zone. When **Balanus** was removed, **Chthamalus** spread down into the freed space.
- **The one-line framing:** species that compete either **exclude** each other or **divide** the resources — there is no third ending.

COMPETITIVE EXCLUSION — ONE STREET, TWO DOSA STALLS

Two stalls open on the same tiny street selling identical dosas at identical prices. Customers split until one stall starves — **competitive exclusion**. If instead one stall switches to idli, both live — **resource partitioning**. If the dosa stall closes on vacation and the idli stall quickly starts selling dosas too, that is **competitive release**. The mapping: customers are the shared resource, stalls are species, and the menu is the niche.

WORKED EXAMPLE — NAME THE PRINCIPLE

One line per scene, three possible answers: exclusion, partitioning, release.

- **Two parakeet species eat the same figs in the same fig tree; after a few seasons only one species remains. Competitive exclusion** — one niche, one winner, the Gause ending.
- **Two woodpecker species share one forest: one drills into dead trunks, the other probes living bark. Resource partitioning** — the niche is split, so both persist.
- **Goats introduced on an island push a small lizard to the rocky coast only; cull the goats, and the lizard spreads back across the island. Competitive release** — the competitor left, so the range expanded.

Test each scene with one question: **did the two species share the whole niche, split it, or lose the competitor?** The answer names the principle before you even read the options.

Situation	Principle at work	Classic study
Two species, one identical niche, one habitat	Competitive exclusion — the weaker is eliminated	Gause's two **Paramecium** species in one tube
Five birds, one tree, different feeding zones	Resource partitioning — niche divided, all persist	MacArthur's five warbler species
Dominant competitor removed, loser expands	Competitive release — range stretches into freed space	Connell's barnacles: **Balanus** vs **Chthamalus**

Situation	Principle at work	Classic study
Weed choked farmland, a moth was set loose	Biological control by a predator	Prickly pear vs **Cacto-blastis** moth, Australia

Interactions II: Parasitism, Commensalism, Mutualism, Amensalism

The remaining four interactions look gentler — some are. Learn each with its sign pair and one example; the summary table at the end of the section is the page to revise from.

Parasitism: +/−

- **The deal:** the parasite (+) lives on or in the host (−) and takes nourishment — harming but not instantly killing, since a dead host ends the free meal.
- **Ectoparasites — outside workers:** lice on scalps, ticks on dogs — they feed from the body surface.
- **Endoparasites — inside workers:** the tapeworm in the intestine, **Plasmodium** in the blood (the malaria parasite).
- **Complex life cycles — host switching:** many parasites must move between hosts to complete their lives. **Plasmodium** needs a **mosquito and a human**; the liver fluke shuttles between a **snail and a sheep**. One host, one stage; another host, the next.
- **Brood parasitism:** the cuckoo (+) lays its eggs in a crow's nest (−) and leaves; the crow incubates the eggs and feeds the chick — unpaid parenting, with the cuckoo chick often pushing the crow's own chicks out.

Commensalism: +/0

- **The deal:** one species gains, the other is **unaffected** — neither helped nor harmed.
- **Cattle egret and cattle:** as cattle graze, insects jump out of the grass and the egrets snap them up. The cattle walk on, indifferent.
- **Clownfish and sea anemone:** the clownfish shelters among the anemone's stinging tentacles, safe where other fish die; the anemone gains nothing.
- **Barnacles on a whale:** the barnacles hitch a free ride through food-rich water; the whale swims on, unbothered.

Mutualism: +/+

- **The deal:** both species gain, and in some pairs the gain becomes a **lifeline** — neither can complete its life without the other.
- **Lichen:** a **fungus and an alga** living as one organism — the alga makes food by photosynthesis, the fungus wraps it in shelter and minerals. Neither grows alone in harsh rock habitats.
- **Mycorrhiza:** fungi wrapped around plant roots (in pines, orchids) that pull in water and minerals; the plant pays in sugars.
- **Fig and wasp — the obligate pair:** the fig fruit can ripen only if a wasp pollinates it; the wasp can lay eggs only inside a fig. Remove either partner and both lines stop.

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- **Ophrys — the sexual deceit:** the Mediterranean orchid **Ophrys** shapes and scents one petal like a **female bee**. Male bees attempt to mate with the flower and fly off carrying pollen. The bee gains nothing; the orchid gets pollinated — a mutualism with a con at its centre.

Amensalism: −/0

- **The deal:** one species is harmed, the other is **unaffected** — the causer gains nothing from the damage it causes.
- **Antibiosis:** the mould **Penicillium** leaks penicillin, which kills bacteria nearby — the mould neither eats nor needs the dead bacteria; it simply poisons its neighbourhood.
- **The sign-check that saves marks:** amensalism is −/0, never −/ +. If the causer **benefits**, you are looking at parasitism or predation instead.

The six interactions on one table

Interaction	Species A	Species B	Classic example
Mutualism	+	+	Lichen (alga + fungus); fig and wasp; mycorrhiza
Predation	+	−	Tiger takes a deer; **Cactoblastis** caterpillars eat prickly pear
Parasitism	+	−	Lice and ticks (ecto); tapeworm and **Plasmodium** (endo); cuckoo's brood parasitism
Commensalism	+	0	Cattle egret and cattle; clownfish and sea anemone; barnacles on a whale
Amensalism	−	0	**Penicillium** mould's penicillin killing nearby bacteria (antibiosis)
Competition	−	−	Gause's **Paramecium**; two barnacles on one rock face

Note the honest wrinkle: predation and parasitism carry the **same** signs (+/ −). The difference is scale — a predator kills quickly and eats whole; a parasite draws slowly from a living host. Read the sentence in the question and choose accordingly.

Putting It Together

Ecology is one long story told in four moves. Walk it once more, slowly, and the whole unit holds together in a single line of thinking.

- **Move 1 — the squeeze:** four abiotic dials (temperature, water, light, soil) set the rules of every habitat. Sub-zero poles, 50 °C deserts, 35‰ seas — someone lives in each, but only the right bodies.

- **Move 2 — the answer:** each organism copes in one of four ways — **regulate** (hold the inside steady), **conform** (follow the outside), **migrate** (leave for a season), **suspend** (pause life). Adaptations are the long-term body versions of the same four answers.
- **Move 3 — the numbers:** a population keeps score with per-capita birth and death rates, sex ratio and age pyramids, then moves along one of two curves — exponential J (no limits) or logistic S (capped at K, fastest at K/2).
- **Move 4 — the neighbours:** species interact in exactly six ways — mutualism, predation, parasitism, commensalism, amensalism, competition — and each pair of signs (+/+, +/−, +/0, −/0, −/−) names its owner.

How to think about an ecology MCQ

Run this three-question checklist before touching the options:

1. **Which interaction?** Assign +, −, 0 to both species. Two + signs → mutualism; +/− with quick killing → predation, slow draining → parasitism; +/0 → commensalism; −/0 → amensalism; −/− → competition.
2. **Which response?** Ask "is the inside steady?" — yes → regulate; no → moving (migrate), paused (suspend) or just following (conform).
3. **Which curve?** Mention of unlimited resources or J-shape → exponential; mention of a ceiling, resources running out or fastest-at-half-K → logistic.

THE WHOLE UNIT IN ONE HOTEL STORY

A hotel sits in a harsh hill station — the **climate is the abiotic squeeze** (Move 1). Every guest copes differently: some switch on their room heater (**regulate**), some just wear whatever the corridor feels like (**conform**), some check out for the worst month (**migrate**), and some simply sleep through it (**suspend**) — that is Move 2. The reception desk keeps the hotel's report card: arrivals, departures, the guest mix by age (**attributes**) and whether the guest count is climbing toward full (**exponential**) or settling at full house (**logistic**, K = every bed taken) — Move 3. And in the dining hall, guests strike deals over the buffet — sharing, stealing, freeloading, ignoring — the six interactions, Move 4. One building, one chapter.

The two curves, side by side

Feature	Exponential (J)	Logistic (S)
Assumption	Resources unlimited.	Resources finite; K caps the habitat.
Equation	$\frac{dN_t}{dt} = rN_t$	$\frac{dN_t}{dt} = rN_t(K - N_t)/K$
Final shape	Rises forever in theory — crashes in reality.	Flattens exactly at K, and holds there.
Fastest growth	As late as the curve runs — no internal peak.	At $N = K/2$, the halfway mark.

Feature	Exponential (J)	Logistic (S)
Typical user	r-strategist boom in a fresh habitat.	K-strategist population near a crowded ceiling.

Response or adaptation? A 10-second judge

- **Sweating at noon:** a **response** — the body's hourly move.
- **Longer limbs in a hot-climate fox:** an **adaptation** — a body feature built over generations (Allen's rule, in reverse for heat).
- **A snail sleeping through summer:** a **response** — suspension, hour by hour.
- **Night-opening stomata in a desert plant:** an **adaptation** — the CAM pattern is written into the species.

Ask yourself	If yes, it is a...	If no, it is a...
Is the inside state held steady against the outside?	Regulator (homeostasis, energy-hungry)	Conformer (inside follows outside)
Is the individual itself moving to kinder conditions?	Migrator — a **response**, temporary and returning	Suspended or conformer — life paused or endured in place
Was the trait built into the body over generations?	Adaptation — the long-term answer	Response — the hourly answer

Spot-the-interaction drill

Scene	Interaction
A honeybee collects nectar and carries pollen for the flower	Mutualism (+/+)
A tick feeds on a dog's blood for weeks	Parasitism (+/−)
An orchid fish guards its home inside stinging tentacles, its host unmoved	Commensalism (+/0)
A Penicillium patch sterilises the bacteria around it	Amensalism (−/0)
Two seedlings race for the same patch of light and both grow weaker	Competition (−/−)
A leopard kills a chital deer in one attack	Predation (+/−)

Three numbers worth carrying out of this chapter: salinity **5‰ inland vs 30–35‰** in the sea, the exponential multiplier $\mathbf{e = 2.718}$, and the logistic fast-point $\mathbf{K/2}$. The full fact list waits in the cheatsheet below.

OurMocks

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET

Term / Fact	Meaning	Example / number
Eurythermal	Tolerates a wide temperature range.	Humans, most mammals
Stenothermal	Tolerates only a narrow temperature band.	Corals, polar fish
Salinity	Dissolved salt, in parts per thousand (‰).	Inland ≈ 5‰; sea 30–35‰
Halophiles	Organisms that tolerate salty water.	Life in salt pans and brine
Photoperiod	Length of daylight — the plant clock.	Flowering on schedule; migration cues
Regulate	Hold the inside steady (homeostasis).	Humans sweat and shiver
Conform	Body follows the outside.	Most invertebrates
Migrate	Move away for a season, then return.	Siberian birds → Bharatpur
Suspend	Pause life until the season turns.	Hibernation (winter), aestivation (summer), diapause (insects)
Per-capita rate	Births or deaths ÷ population size.	20 births in 1,000 → 0.02
Age pyramid — growing	Wide base: many pre-reproductive young.	Boom ahead
Age pyramid — stable	Even sides across all three age groups.	Population holds level
Age pyramid — declining	Narrow base: fewer young than parents.	Shrink ahead
Size without counting	% cover or biomass when individuals cannot be counted.	Banyan tree cover; grass biomass
Exponential rate	$\frac{dN_t}{dt} = rN_t$ — growth speeds up with crowd size.	J-curve; r = per-capita rate
Exponential size	$N_t = N_0 \times e^{rt}$ — size after time t.	N_0 start; $e = 2.718$; t = time
Logistic rate	$\frac{dN_t}{dt} = rN_t(K - N_t)/K$ — exponential times free resources.	S-curve; $(K - N_t)/K$ = free fraction

Term / Fact	Meaning	Example / number
K (carrying capacity)	The largest population the habitat's resources can support.	Growth stops at K
Fastest growth	At $N = K/2$ — many breeders and resources still free.	Not at K, not at 0
r-strategist	Many offspring, no parental care.	Insects, small fish runs
K-strategist	Few offspring, heavy parental care.	Elephants, large mammals
Life history variation	Breed once vs many; age at first breeding; clutch size bet.	Bamboo flowers once; fish spawn thousands
Predation	Predator kills and eats (+/ −).	Tiger–deer; Cactoblastis on cactus
Competition	Both lose over shared resources (−/ −).	Gause's Paramecium: same niche → one eliminated
Resource partitioning	Species split the niche and coexist.	Five warbler species, one tree, five zones
Competitive release	Competitor removed → loser's range expands.	Connell's barnacles: Balanus removed, Chthamalus spread
Parasitism	Parasite lives on/in host (+/ −).	Lice, ticks (ecto); tapeworm, Plasmodium (endo)
Brood parasitism	Eggs dumped in another species' nest.	Cuckoo chick raised by a crow
Commensalism	One gains, other unaffected (+/0).	Cattle egret; clownfish; barnacles on whale
Mutualism	Both gain (+/ +).	Lichen; mycorrhiza; fig–wasp; Ophrys' bee deceit
Amensalism	One harmed, causer unaffected (−/0).	Penicillium's penicillin vs bacteria (antibiosis)

Common Errors and Special Cases

Seven classic confusions fill this table — the middle column is the version worth memorising, and the right column tells you why the wrong version feels so tempting.

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	One-line why / trap
Saying conformers maintain homeostasis	Regulators maintain homeostasis; conformers follow the outside	Homeostasis costs energy — conformers skip that bill
Putting K inside exponential growth	Exponential ignores K; only logistic carries the $(K - N_t)/K$ brake	The J-curve has no ceiling — that is its whole story
Placing fastest logistic growth at K	Fastest at $N = K/2$; growth is **zero** at K	At K the resources are gone, so nothing grows
Marking commensalism as $+/-$ or amensalism as $-/+$	Commensalism $= +/0$; amensalism $= -/0$	The second species must show exactly 0 — no gain, no harm
Expecting every competition to end in exclusion	Exclusion needs the **same** niche; different zones allow coexistence	Partitioning the niche is the escape route (warblers, barnacles)
Calling hibernation a form of migration	Hibernation is a **suspend** response	Life pauses in place — no travel happens
Labelling cuckoo–crow as commensalism	It is brood parasitism, a $+/-$ deal	The crow pays: it raises a stranger while its own chicks lose

Three checks that catch these errors

- **Check the second species first:** in $+/0$ and $-/0$ deals the second partner shows exactly zero. If you wrote $+$ or $-$ for it, you have drifted into parasitism without noticing.
- **Trace both signs before naming:** predation and parasitism share $+/-$ — kill-fast vs drain-slow decides which word to choose.
- **Anchor K-questions on the curve:** fastest at $K/2$, zero at K, brake term $(K - N_t)/K$ — one line of curve-reading covers every logistic arithmetic question.
- **Separate response from adaptation:** responses are hourly moves (sweat, migrate, sleep); adaptations are generational features (cuticle, Allen's rule). The question's tense tells you which one it wants.

The Whole Chapter on One Page

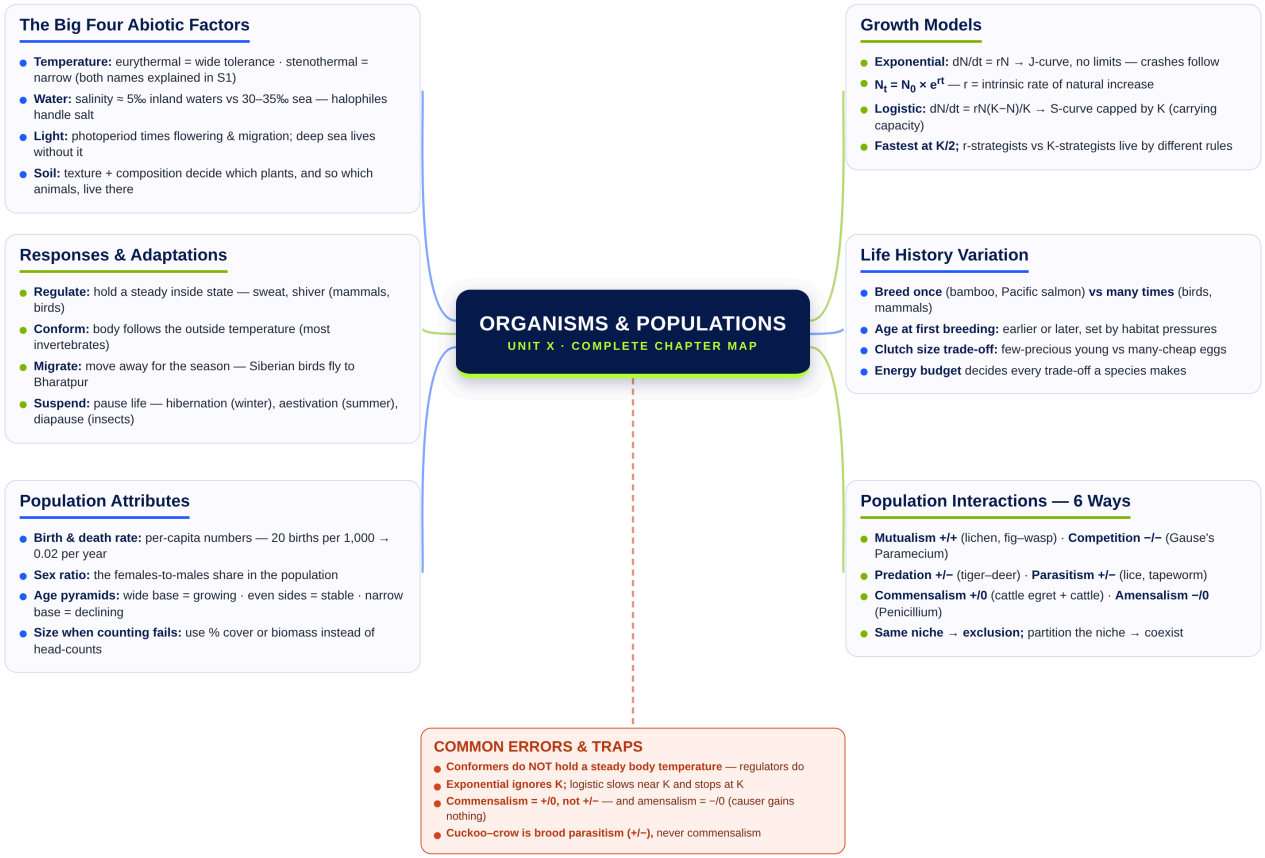


Figure 2: Whole-chapter mindmap — the abiotic dials, the four responses, the population numbers, both growth curves, life-history bets, all six interactions, and the traps that catch most students.

Practice Questions

Q1. Organisms that can tolerate only a **narrow** range of temperatures are called:

(A) Stenothermal
(B) Eurythermal
(C) Euryhaline
(D) Stenohaline

Q2. A culture of 1,000 organisms grows exponentially with $r = 0.2$ per unit time. Using $e^2 = 7.389$, the population after 10 time units is closest to:

(A) 2,718
(B) 7,389
(C) 2,000
(D) 738

Q3. An age pyramid shows a **narrow base** and a broad middle. The population is:

(A) Growing
(B) Stable
(C) Fluctuating
(D) Declining

Q4. The mould Penicillium releases penicillin that kills bacteria around it, while the mould itself gains nothing from those deaths. This interaction is:

(A) Amensalism **(B)** Commensalism
(C) Mutualism **(D)** Parasitism

Q5. A population has $N = 600$ in a habitat with carrying capacity $K = 1,200$. The logistic brake term $(K - N)/K$ equals:

(A) 1.0 **(B)** 2.0
(C) 0.5 **(D)** 0.25

Q6. In logistic growth, the population adds new individuals **fastest** when:

(A) At $N = K$ **(B)** At $N = K/2$
(C) Just below K **(D)** At $N = 0$

Q7. Gause grew two Paramecium species together in the same tube, sharing one niche. The outcome was:

(A) Both species coexisted **(B)** Both species grew faster
(C) One species eliminated the other **(D)** Neither species survived

Q8. Cattle egrets feed on insects that jump up as cattle graze; the cattle gain nothing and lose nothing. This is:

(A) Commensalism **(B)** Mutualism
(C) Parasitism **(D)** Amensalism

Q9. The cuckoo lays its eggs in a crow's nest, and the crow raises the cuckoo chick. This interaction is:

(A) Mutualism **(B)** Brood parasitism
(C) Commensalism **(D)** Predation

Q10. The Ophrys orchid's petal mimics a female bee, so male bees visit and carry away pollen. The orchid's relationship with the bee is:

(A) Commensalism **(B)** Amensalism
(C) Brood parasitism **(D)** Mutualism

Solutions

Sol. Q1. "Steno" means narrow — a **stenothermal** organism tolerates a narrow temperature band (corals, polar fish). "Eury" means wide; and the **-haline** pair is about **salinity**, not temperature.

Answer: (A)

Sol. Q2. Model: $N_t = N_0 e^{rt}$. Exponent: $r \times t = 0.2 \times 10 = 2$. So $N_t = 1,000 \times e^2$, and $1,000 \times 7.389 \approx$ **7,389**. (Choosing $e^1 = 2.718$ gives 2,718 — the classic slip.)

Answer: (B)

Sol. Q3. The base holds pre-reproductive individuals — tomorrow's breeders. A narrow base means fewer young than parents, so births lag behind deaths: the population is **declining**. A wide base marks the growing shape.

Answer: (D)

Sol. Q4. Signs: bacteria are harmed (−), Penicillium is unaffected (0). A −/0 pair is **amensalism** — here via antibiosis. Commensalism would need the mould to gain (+/0).

Answer: (A)

Sol. Q5. Substitute: $(K - N)/K = (1,200 - 600)/1,200 = 600/1,200$, which is **0.5** — half the resources remain free, and $N = 600$ sits exactly at $K/2$, the fastest-growth point.

Answer: (C)

Sol. Q6. Fast growth needs **many breeders** (large N) **and free resources** (large $(K - N)/K$). Both conditions peak together at $N = K/2$. At K growth has stopped; at 0 there are almost no breeders.

Answer: (B)

Sol. Q7. Same niche, same tube — the competitive exclusion principle applies. **P. aurelia** outcompeted and eliminated **P. caudatum**; each species had thrived when grown alone.

Answer: (C)

Sol. Q8. Egret gains food (+); cattle are untouched (0). A +/0 pair is **commensalism**. For mutualism the cattle would have to gain something too.

Answer: (A)

Sol. Q9. Cuckoo gains (+) — a crow does the parenting. Crow loses (−) — it raises a stranger while its own chicks suffer. A +/ − deal with an egg in another's nest is **brood parasitism**.

Answer: (B)

Sol. Q10. Both partners gain (+/ +): the bee's mating attempt serves the orchid's pollination, and over generations the orchid's survival leans on the bee — **mutualism**, built on sexual deceit.

Answer: (D)

Last-mile recall drill

Cover the right-hand column, say the answer aloud, then check yourself:

Prompt	Answer to recall
Wide vs narrow temperature tolerance?	Eurythermal vs stenothermal
The four responses — fight, give in, leave, pause?	Regulate · conform · migrate · suspend
Exponential size after time t, and its constant?	$N_t = N_0 \times e^{rt}$; $e = 2.718$
Where logistic growth runs fastest — and where it stops?	Fastest at $K/2$; stops at K
Same niche, one species eliminated — who proved it?	Gause's Paramecium experiment
The sign pairs of the six interactions?	Mutualism $+/+$ · predation $+/-$ · parasitism $+/-$ · commensalism $+/0$ · amensalism $-/0$ · competition $-/-$
One orchid that cons a bee, and one cuckoo trick?	Ophrys sexual deceit; brood parasitism
Birth rate of 20 births in a 1,000-strong population?	0.02 per year — divide by population size
Moth caterpillars that crushed the prickly pear cactus?	Biological control — Cactoblastis in Australia

Closing habit: say the sign pair out loud before reading the options; place every logistic question at $K/2$ before touching the arithmetic; and match each example to its named study — Gause, Connell, MacArthur — because name-pair questions reward exactly that habit.

Quick-fire judge: true or false?

Statement	Verdict and the fix
Conformers keep their body temperature constant.	False — **regulators** do; a conformer's body follows the outside.
A logistic population can keep growing past K.	False — K is the ceiling; growth **stops** at K.
The cuckoo and the crow are mutualism partners.	False — the crow pays; it is **brood parasitism** $(+/-)$.
20 births among 1,000 people is a birth rate of 20.	False — per-capita arithmetic: $20 \div 1{,}000 =$ **0.02**.
Mycorrhizal fungi help roots absorb water and minerals.	True — the fungus trades minerals for the plant's sugars.

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